

Practical No. 9

AIM: Performing one-way ANOVA using aov() (R).

Input :

```
# =====  
# PRACTICAL 9 – CHI SQUARE TEST  
# Dataset : flipkard.csv  
# =====  
  
df <- read.csv("flipkard.csv", na.strings = c("", "NA", " "))  
  
# -----  
# 1. remove identifier columns automatically  
# -----  
bad_cols <- c("id","ID","product_id","order_id","user_id","serial","sr_no")  
df <- df[, !(names(df) %in% bad_cols)]  
  
# -----  
# 2. detect categorical columns  
# -----  
cat_cols <- names(df)[sapply(df, is.character) | sapply(df, is.factor)]  
  
# if not enough categorical columns, create them  
if(length(cat_cols) < 2){  
  
  num_cols <- names(df)[sapply(df, is.numeric)]  
  
  # create two categorical groups from numeric data  
  df$Group1 <- cut(df[[num_cols[1]]], breaks = 3,  
    labels = c("Low","Medium","High"))  
  
  df$Group2 <- cut(df[[num_cols[1]]], breaks = 2,  
    labels = c("Small","Large"))  
  
  var1 <- "Group1"  
  var2 <- "Group2"  
  
} else {  
  
  # choose two categorical with small number of levels  
  small <- sapply(df[cat_cols], function(x) length(unique(x)))
```

```

small <- sort(small)

var1 <- names(small)[1]
var2 <- names(small)[2]

df[[var1]] <- as.factor(df[[var1]])
df[[var2]] <- as.factor(df[[var2]])
}

# -----
# 3. create contingency table safely
# -----
tab <- table(df[[var1]], df[[var2]])

# remove empty rows/columns
tab <- tab[rowSums(tab) > 0, colSums(tab) > 0]

# -----
# 4. perform chi-square test
# -----
chi_result <- chisq.test(tab)

cat("\nVARIABLE 1:", var1)
cat("\nVARIABLE 2:", var2, "\n")

cat("\n--- CONTINGENCY TABLE ---\n")
print(tab)

cat("\n--- CHI-SQUARE RESULT ---\n")
print(chi_result)

# p-value only
cat("\nP-VALUE =", chi_result$p.value, "\n")
# Interpretation
if(chi_result$p.value < 0.05){
  cat("\nConclusion: Significant association → REJECT H0\n")
} else {
  cat("\nConclusion: No significant association → ACCEPT H0\n")
}

```

Output :

```
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Source
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> df <- read.csv("flipkart.csv", na.strings = c("", "NA", " "))
> # -----
> # 1. remove identifier columns automatically
> bad_cols <- c("id", "ip", "product_id", "order_id", "user_id", "serial", "sr_no")
> df <- df[, ! (names(df) %in% bad_cols)]
> # -----
> # 2. detect categorical columns
> cat_cols <- names(df)[apply(df, is.character) | apply(df, is.factor)]
> # if not enough categorical columns, create them
> if (length(cat_cols) < 2) {
+   num_cols <- names(df)[apply(df, is.numeric)]
+   # create two categorical groups from numeric data
+   df$group1 <- cut(df[[num_cols[1]]], breaks = 3,
+     labels = c("Low", "Medium", "High"))
+   df$group2 <- cut(df[[num_cols[1]]], breaks = 2,
+     labels = c("Small", "Large"))
+   var1 <- "Group1"
+   var2 <- "Group2"
+ } else {
+   # choose two categorical with small number of levels
+   small <- sapply(df[cat_cols], function(x) length(unique(x)))
+   small <- sort(small)
+   var1 <- names(small)[1]
+   var2 <- names(small)[2]
+   df[[var1]] <- as.factor(df[[var1]])
+   df[[var2]] <- as.factor(df[[var2]])
+ }
> # -----
> # 3. create contingency table safely
> # -----
> tab <- table(df[[var1]], df[[var2]])
> # remove empty rows/columns
> tab <- tab[rowSums(tab) > 0, colSums(tab) > 0]
```

Environment History Connections Tutorial

R - Global Environment

- chi_result List of 9
- df 80000 obs. of 24 variables
- flipkart 80000 obs. of 25 variables
- model List of 14

Values

bad_cols	chr [1:7]	"id" "ip" "product_id" "order..."
cat_cols	chr [1:10]	"product_name" "category" "b..."
formula_used	AT ~ Group	
group_var	"Group"	
num_cols	chr [1:5]	"AT" "V" "AP" "RH" "PE"
numeric_var	"AT"	
pvals	num [1:3]	4.0e-65 7.1e-29 NA
small	Named int [1:10]	2 4 6 8 8 8 8 15 2001 ...

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```
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Source
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> # -----
> chi_result <- chisq.test(tab)
> cat("\nVARIABLE 1:", var1)

VARIABLE 1: is_returnable
> cat("\nVARIABLE 2:", var2, "\n")

VARIABLE 2: payment_modes
> cat("\n--- CONTINGENCY TABLE ---\n")

--- CONTINGENCY TABLE ---
> print(tab)

      CARD,Wallet COD,CARD COD,UPI,CARD UPI,CARD
False    4002    3864      4075     4025
True     16112   16008     15902    16012
> cat("\n--- CHI-SQUARE RESULT ---\n")

--- CHI-SQUARE RESULT ---
> print(chi_result)

Pearson's Chi-squared test

data:  tab
X-squared = 5.9659, df = 3, p-value = 0.1133
> # p-value only
> cat("\np-VALUE =", chi_result$p.value, "\n")

P-VALUE = 0.1132801
> # Interpretation
> if (chi_result$p.value < 0.05) {
+   cat("\nConclusion: Significant association - REJECT H0\n")
+ } else {
+   cat("\nConclusion: No significant association - ACCEPT H0\n")
+ }

Conclusion: No significant association - ACCEPT H0
>
>
>
> |
```

Environment History Connections Tutorial

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Values

bad_cols	chr [1:7]	"id" "ip" "product_id" "order..."
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numeric_var	"AT"	
pvals	num [1:3]	4.0e-65 7.1e-29 NA
small	Named int [1:10]	2 4 6 8 8 8 8 15 2001 ...

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